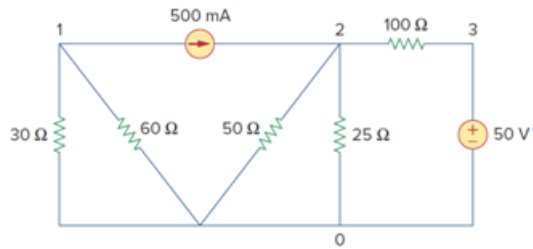


Problem

For the circuit in Fig. 3.33, use *PSpice* to find the node voltages.

Figure 3.33:



Step-by-step solution

Step 1 of 7

Consider the following circuit diagram to find the node voltages:

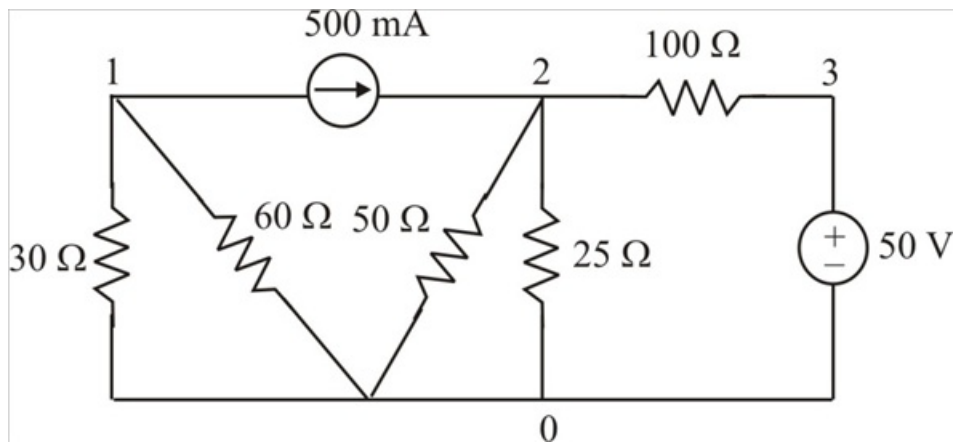


Figure 1

Step 2 of 7

The $30\ \Omega$ and $60\ \Omega$ resistances in the circuit of Figure 1 are in parallel.

Calculate their combined resistance:

$$\begin{aligned}(30\ \Omega \parallel 60\ \Omega) &= \frac{(30\ \Omega)(60\ \Omega)}{30\ \Omega + 60\ \Omega} \\ &= \frac{1800}{90}\ \Omega \\ &= 20\ \Omega\end{aligned}$$

The $50\ \Omega$ and $25\ \Omega$ resistances in the circuit of Figure 1 are in parallel.

Calculate their combined resistance:

$$\begin{aligned}(50\ \Omega \parallel 25\ \Omega) &= \frac{(50\ \Omega)(25\ \Omega)}{50\ \Omega + 25\ \Omega} \\ &= \frac{1250}{75}\ \Omega \\ &= 16.67\ \Omega\end{aligned}$$

Step 3 of 7

Label the circuit in Figure 1 and redraw as shown in Figure 2:

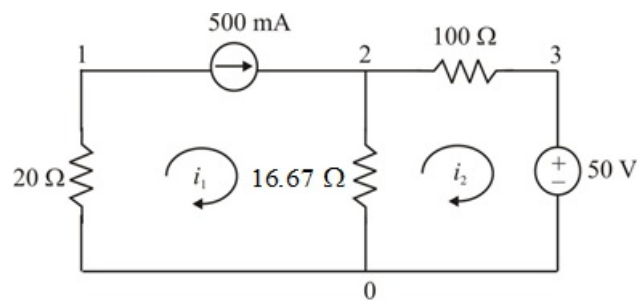


Figure 2

Step 4 of 7

Kirchhoff's voltage law states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Apply Kirchhoff's voltage law to the loop 1 of circuit in Figure 2 to get the current i_1 as follows:

$$\begin{aligned} i_1 &= 500 \text{ mA} \\ &= 0.5 \text{ A} \end{aligned}$$

Calculate the voltage v_1 across the 20Ω resistance:

$$\begin{aligned} v_1 &= -i_1 R \\ &= -(0.5 \text{ A})(20 \Omega) \\ &= -10 \text{ V} \end{aligned}$$

Therefore, the voltage at node 1, v_1 is -10 V.

Step 5 of 7

Apply Kirchhoff's voltage law to the loop 2 of circuit in Figure 2 to get the current i_2 as follows:

$$(16.67 \, \Omega)(i_2 - i_1) + (100 \, \Omega)i_2 + 50 \, \text{V} = 0$$

$$(16.67 \, \Omega)i_2 - (16.67 \, \Omega)i_1 + (100 \, \Omega)i_2 + 50 \, \text{V} = 0$$

$$(116.67 \, \Omega)i_2 - (16.67 \, \Omega)i_1 + 50 \, \text{V} = 0$$

Substitute 0.5 A for i_1 and simplify the expression.

$$(116.67 \, \Omega)i_2 - (16.67 \, \Omega)(0.5 \, \text{A}) + 50 \, \text{V} = 0$$

$$(116.67 \, \Omega)i_2 - 8.34 \, \text{V} + 50 \, \text{V} = 0$$

$$(116.67 \, \Omega)i_2 = -41.67 \, \text{V}$$

$$i_2 = \frac{-41.67 \, \text{V}}{116.67 \, \Omega}$$

$$= -0.357 \, \text{A}$$

Calculate the current i_o flowing through the $16.67 \, \Omega$ resistance.

$$i_o = i_1 - i_2$$

$$= 0.5 \, \text{A} - (-0.357 \, \text{A})$$

$$= 0.857 \, \text{A}$$

Calculate the voltage v_2 across the $16.67 \, \Omega$ resistance:

$$v_2 = i_o R$$

$$= (0.857 \, \text{A})(16.67 \, \Omega)$$

$$= 14.286 \, \text{V}$$

Therefore, the voltage at node 2, v_2 is 14.286 V.

From the circuit in Figure 1, it is clear that the voltage v_3 at node 3 is equal to the 50 V voltage source. Therefore,

$$v_3 = 50 \, \text{V}$$

Therefore, the voltage at node 3, v_3 is 50 V.

Step 6 of 7

PSPICE Analysis:

Use the PSPICE schematics to draw the circuit as shown in Figure 3:

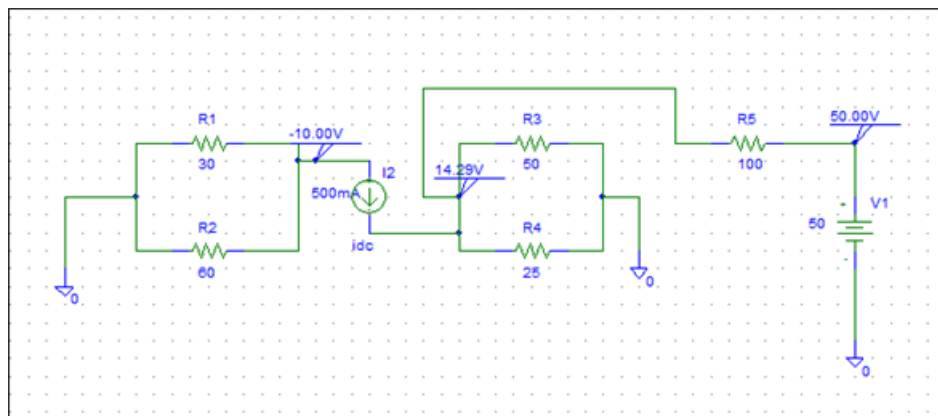


Figure 3

Step 7 of 7

Notice that the positive terminal of the 50 V voltage source is connected to the inverting terminal (pin 2) via resistors, while non-inverting terminal is connected to pin 3, also notice that the positive power supply terminal (pin 4) is connected to 15 V dc voltage source, while the negative power supply terminal (pin 11) is connected to -15 V voltage source. Add a VIEWPOINT to measure the following node voltages.

VIEWPOINT reads -10 V , 14.29 V, and 50 V.

Compare these values with the theoretical values -10 V , 14.286 V, and 50 V respectively.

A simplified analysis has produced the same result as computer analysis as shown in Figure 3.